MERN STACK

DAY – 19 16 July 2026

React

useContext

useContext is a built-in React Hook that allows a component to read and use data from a Context. It provides a way to share data across multiple components without passing props through every intermediate component.

It is commonly used for global or shared state that many components need to access.

Why useContext is Needed

In React, data normally flows from a parent component to a child component through props. When the same data is required by many nested components, the data has to be passed through every level of the component tree.

This process is called prop drilling.

useContext solves this problem by allowing components to access shared data directly from a Context, regardless of how deeply they are nested.

Context

Context is a feature in React that creates a central place to store and share data.

A Context consists of three parts:

1. Context Object – Created using createContext().

```
import { createContext } from "react";
const MyContext = createContext();
```

2. Provider – Supplies the shared value to components.

```
<MyContext.Provider value={value}>
  <Child />
</MyContext.Provider>
```

3. Consumer (useContext) – Reads the shared value.
4.

const value = useContext(ContextName);

TASK -1

```
import { createContext } from "react";

export const Createcon=createContext(null);
```

```
import logo from './logo.svg';
import './App.css';
import { Day1 } from './components/day1';
import {Header} from './components/header';
import { Routess } from './components/routes';
import { Us } from './components/counter.js';
import { Todo } from './components/todo';
import { Ue } from './components/USEEFFECT.js';
import { Createcon } from './components/createcon.js';
import { useState } from 'react';
import { Usecon } from './components/usecon.js';
import { Usecon2 } from './components/usecon2.js';
function App() {
  const [name,setName]=useState("")
  return (
    <div className="App">

     <Createcon.Provider value={[name,setName]}>
      <Usecon/>
      <Usecon2/>
     </Createcon.Provider>
    </div>
  );
}

export default App;
```

```
import { useContext } from "react"
import { Createcon } from "./createcon";

export const Usecon=()=>{
    const [name,setName]=useContext(Createcon)
    return (

        <>
        <input type="text" onChange={(e)=>setName(e.target.value)}/>
        {name} This is 1st Output.
        </>
    );
}
```

```jsx
import { useContext } from "react"
import { Createcon } from "./createcon";

export const Usecon2=()=>{
    const [name,setName]=useContext(Createcon)
    return (

        <>
        <input type="text" onChange={(e)=>setName(e.target.value)}/>
        {name} This is 2nd Output.
        </>
    );
}
```

ANJALI KUMARI | ANJALI KUMARI This is 1st Output. | | ANJALI KUMARI This is 2nd Output.

Local Storage

Local Storage is a browser storage mechanism used to store data permanently as key-value pairs. The data remains even after closing the browser until it is manually removed.

Set Data:

```
localStorage.setItem("key", "value");
```

Get Data:

```
localStorage.getItem("key");
```

Session Storage

Session Storage is a browser storage mechanism used to store data temporarily for a single browser session. The data is removed automatically when the browser tab is closed.

Set Data:

```
sessionStorage.setItem("key", "value");
```

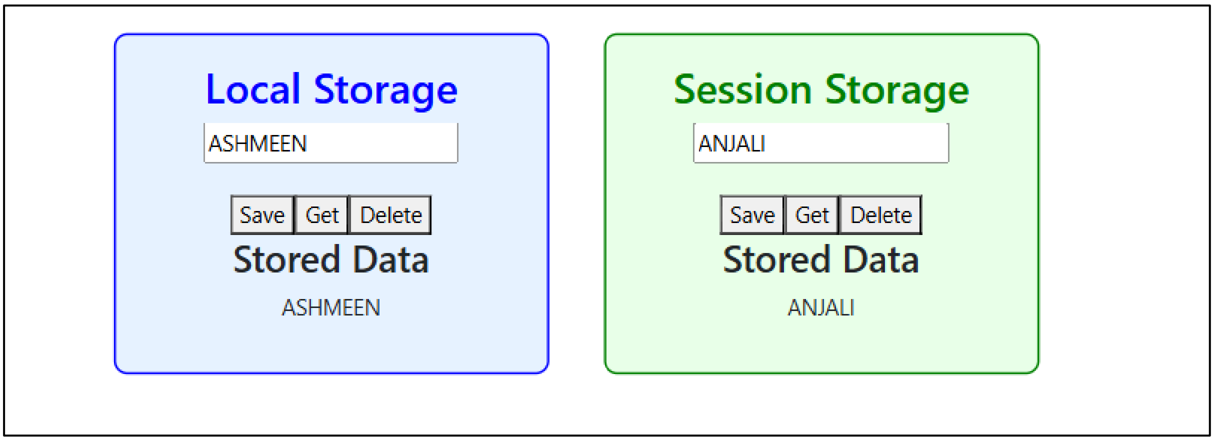
Get Data:

```
sessionStorage.getItem("key");
```

TASK-2

```
import { useState } from "react";
export const Ls = () => {
  const [name, setName] = useState("");
  const [val, setVal] = useState("");
  const [sname, setSname] = useState("");
  const [sval, setSval] = useState("");
  const addLocal = () => {
    localStorage.setItem("Data", JSON.stringify(name));
  };
  const getLocal = () => {
    setVal(JSON.parse(localStorage.getItem("Data")));
  };
  const deleteLocal = () => {
    localStorage.removeItem("Data");
    setVal("");
  };
  const addSession = () => {
    sessionStorage.setItem("SessionData", JSON.stringify(sname));
  };
  const getSession = () => {
    setSval(JSON.parse(sessionStorage.getItem("SessionData")));
  };
  const deleteSession = () => {
    sessionStorage.removeItem("SessionData");
    setSval("");
  };
  return (
    <div style={{ display: "flex", gap: "40px", justifyContent: "center", marginTop: "30px" }}>
      <div
        style={{
          width: "320px",
          padding: "20px",
          border: "2px solid blue",
          borderRadius: "10px",
          backgroundColor: "#e6f2ff",
          textAlign: "center",
        }}
      >
```

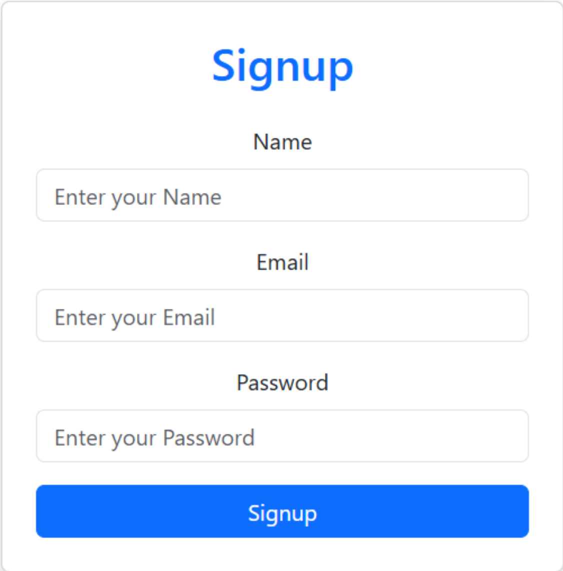
```
        <h2 style={{ color: "blue" }}>Local Storage</h2>
        <input
          type="text"
          value={name}
          onChange={(e) => setName(e.target.value)}
        />
        <br /><br />
        <button onClick={addLocal}>Save</button>
        <button onClick={getLocal}>Get</button>
        <button onClick={deleteLocal}>Delete</button>
        <h3>Stored Data</h3>
        <p>{val}</p>
      </div>
      <div
        style={{
          width: "320px",
          padding: "20px",
          border: "2px solid green",
          borderRadius: "10px",
          backgroundColor: "#e8ffe8",
          textAlign: "center",
        }}
      >
        <h2 style={{ color: "green" }}>Session Storage</h2>
        <input
          type="text"
          value={sname}
          onChange={(e) => setSname(e.target.value)}
        />
        <br /><br />
        <button onClick={addSession}>Save</button>
        <button onClick={getSession}>Get</button>
        <button onClick={deleteSession}>Delete</button>
        <h3>Stored Data</h3>
        <p>{sval}</p>
      </div>
    </div>
  );
};
```

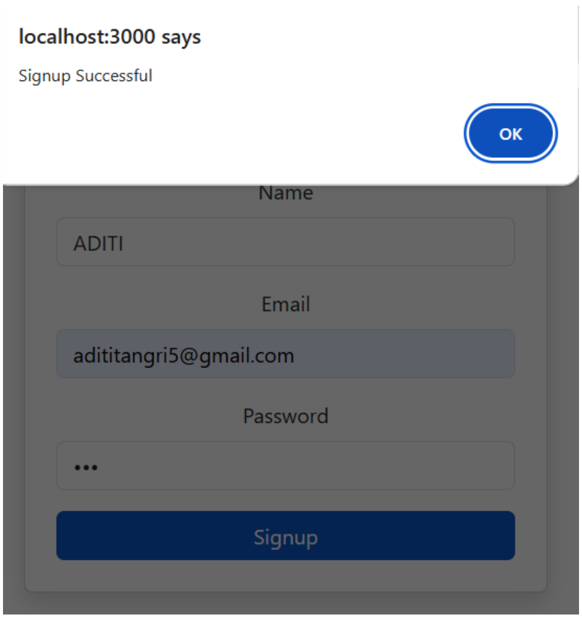


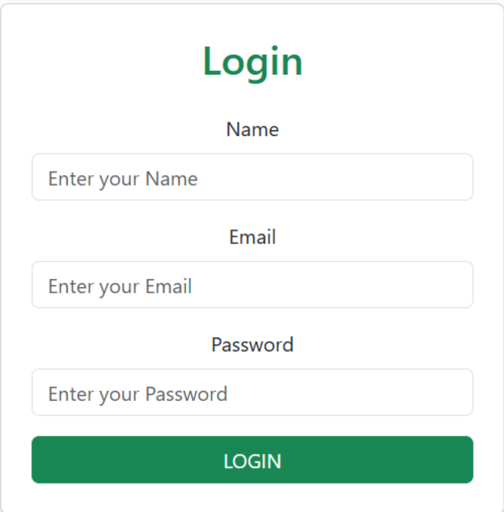
TASK -3 LOGIN SIGNUP PAGE

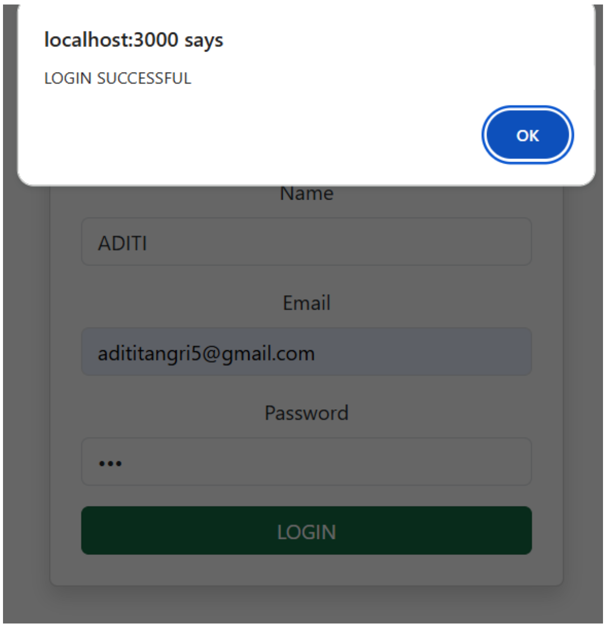
```jsx
import { useState } from "react";

export const Login = () => {
  const [name, setName] = useState("");
  const [emai, setEmai] = useState("");
  const [pass, setPass] = useState("");

  const get1 = () => {
    let n = JSON.parse(localStorage.getItem("Name"));
    let e = JSON.parse(localStorage.getItem("Email"));
    let p = JSON.parse(localStorage.getItem("Password"));

    if (n === name && e === emai && p === pass) {
      alert("LOGIN SUCCESSFUL");
    } else {
      alert("LOGIN FAILED!! INVALID CREDENTIALS");
    }
  };

  return (
    <div className="container mt-5">
      <div className="card shadow p-4 mx-auto" style={{ maxWidth: "400px" }}>

        <h2 className="text-center text-success mb-4">Login</h2>

        <div className="mb-3"><label className="form-label">Name</label><input className="form-control" type="text" placeholder="Enter your Name" onChange={(e)=>setName(e.target.value)} /></div>

        <div className="mb-3"><label className="form-label">Email</label><input className="form-control" type="email" placeholder="Enter your Email" onChange={(e)=>setEmai(e.target.value)} /></div>

        <div className="mb-3"><label className="form-label">Password</label><input className="form-control" type="password" placeholder="Enter your Password" onChange={(e)=>setPass(e.target.value)} /></

        <button className="btn btn-success w-100" onClick={get1}>LOGIN</button>

      </div>
    </div>
  );
};
```

```jsx
import { useState } from "react";
import { useNavigate } from "react-router-dom";

export const Sign = () => {
  const [name, setName] = useState("");
  const [emai, setEmai] = useState("");
  const [pass, setPass] = useState("");

  const navigate = useNavigate();

  const add1 = () => {
    localStorage.setItem("Name", JSON.stringify(name));
    localStorage.setItem("Email", JSON.stringify(emai));
    localStorage.setItem("Password", JSON.stringify(pass));
    alert("Signup Successful");
    navigate("/login");
  };

  return (
    <div className="container mt-5">
      <div className="card shadow p-4 mx-auto" style={{ maxWidth: "400px" }}>
        <h2 className="text-center text-primary mb-4">Signup</h2>

        <div className="mb-3"><label className="form-label">Name</label><input type="text" className="form-control" placeholder="Enter your Name" onChange={(e)=>setName(e.target.value)} /></div>

        <div className="mb-3"><label className="form-label">Email</label><input type="email" className="form-control" placeholder="Enter your Email" onChange={(e)=>setEmai(e.target.value)} /></div>

        <div className="mb-3"><label className="form-label">Password</label><input type="password" className="form-control" placeholder="Enter your Password" onChange={(e)=>setPass(e.target.value)} /></

        <button className="btn btn-primary w-100" onClick={add1}>Signup</button>
      </div>
    </div>
  );
};
```


Signup
Name
Enter your Name
Email
Enter your Email
Password
Enter your Password
Signup


localhost:3000 says
Signup Successful
OK
Name
ADITI
Email
adititangri5@gmail.com
Password
•••
Signup


Login
Name
Enter your Name
Email
Enter your Email
Password
Enter your Password
LOGIN


localhost:3000 says
LOGIN SUCCESSFUL
OK
Name
ADITI
Email
adititangri5@gmail.com
Password
•••
LOGIN